Signal-to-Shock

Financial Event Intelligence & Portfolio Stress Testing

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The Problem

Financial portfolios are exposed to rapid, unstructured events — earnings surprises, geopolitical shocks, credit downgrades, and cyber incidents. Risk teams need a fast path from **news signal** to **quantified portfolio impact**.

Today that gap is bridged manually, slowly, and inconsistently.

Why It Matters

External events create portfolio risk faster than manual processes can respond:

Credit Event (e.g. SVB Collapse)	Geopolitical Shock (e.g. UK Mini-Budget)	Macro Announcement (e.g. Rate Decision)
Bank-run headlines → regional banking ETF KRE fell –19.9% over 5 trading days vs SPY –0.7%	Policy surprise → UK equity EWU fell –5.5% while broad US market fell –4.6%	Systemic re-pricing across equities, bonds, derivatives simultaneously

Different event *types* have different *scopes*: issuer-specific events affect targeted positions; systemic events reprice entire asset classes.

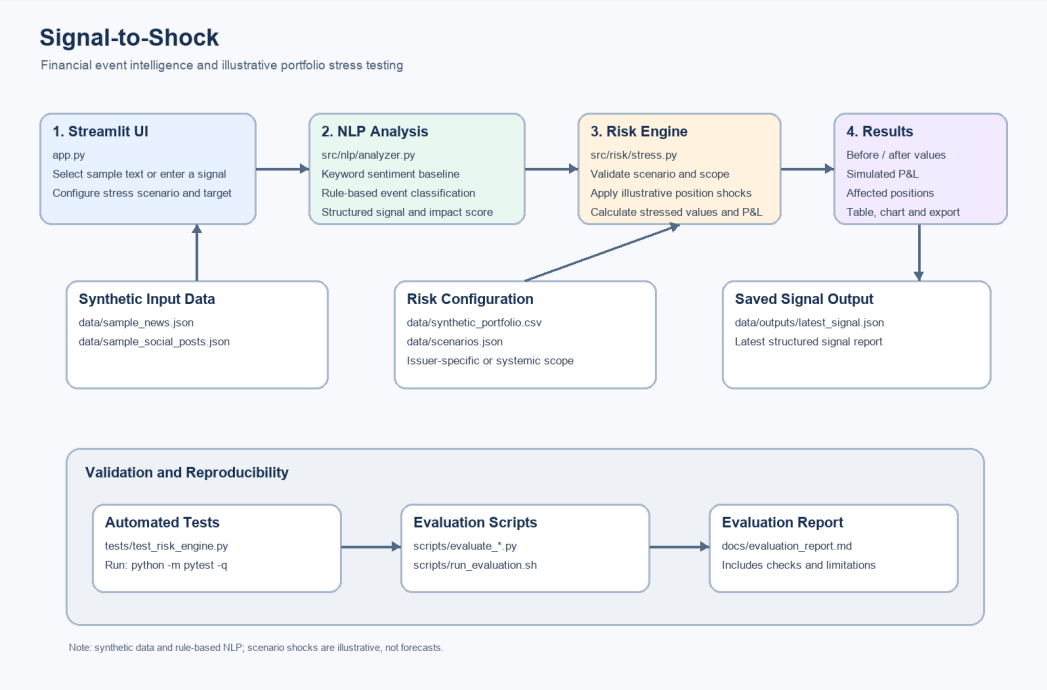
Signal-to-Shock automates the **news** → **classification** → **impact** → **stress test** workflow so risk analysts see quantified exposure in seconds.

■ *Portfolio shock magnitudes in this prototype are **illustrative assumptions**, not calibrated market forecasts.*

Solution & Architecture

① Input	Financial news or social-media text (synthetic demo data)
② Sentiment Score	Keyword rule: (pos_count – neg_count) / (pos_count + neg_count)
③ Event Class	Rule-based: Credit Event, Geopolitical, Macro, M&A;, Cyber, Other
④ Impact Score	0–10 severity from keyword signals
⑤ Stress Test	market_value × (1 + shock × impact/10) [illustrative shocks]
⑥ Output	Before/after P&L;, per-position breakdown, affected flag

36 unit tests · GitHub Actions CI · Python 3.12



Architecture diagram — docs/architecture.png

Step	User Action	System Response
1	Select or type financial news / social-media text	Displayed in the input panel
2	Click Analyse Signal	Sentiment score, event type, impact score (0–10) computed
3	View Signal Panel	Structured output: event class, affected issuer, scenario name
4	Inspect Portfolio	Before/after values, P&L per position, affected positions highlighted
5	Switch event / issuer	Re-run stress with different scenario scope (issuer vs. systemic)

Synthetic Demo Portfolio (6 positions, total $1,150,000)

Asset	Type	Issuer	Market Value
Technology Equity	Equity	Example Technology Ltd	$250,000
Banking Equity	Equity	Example Bank Ltd	$200,000
Corporate Bond	Bond	Example Technology Ltd	$150,000
Bank Loan	Loan	Example Bank Ltd	$250,000
Government Bond	Bond	Example Government	$200,000
Interest Rate Swap	Derivative	Example Bank Ltd	$100,000

■ *Portfolio positions and scenario shock magnitudes are **synthetic and illustrative** — not real financial data or investment advice.*

Demo video: *Pending — URL to be added after upload* · Run locally: `streamlit run app.py`

Dataset

- Source: Financial PhraseBank (Malo et al., 2014)
- Hugging Face: `takala/financial_phrasebank`
- Licence: CC BY-NC-SA 3.0 — evaluation use only, raw text not redistributed
- Config: sentences_75agree (≥75% annotator agreement)
- Size: 3,453 labelled sentences — Neg: 420, Neu: 2,146, Pos: 887

Methodology

- Existing rule-based analyzer applied *without any tuning*
- Score > 0 → positive | Score < 0 → negative | Score = 0 → neutral
- No model training; this is a **baseline evaluation only**

Overall Accuracy: 67.30% | Macro F1: 0.4314

■ *This evaluates only sentiment extraction. It does not validate event classification, impact scoring, or portfolio stress assumptions.*

Per-Class Performance

Class	Precision	Recall	F1
Negative	45.98%	9.52% ■	0.158
Neutral	69.29%	96.83%	0.808
Positive	56.13%	23.22%	0.329

Confusion Matrix

True \ Pred	Neg	Neu	Pos
Neg (420)	40	279	101
Neu (2146)	8	2078	60
Pos (887)	39	642	206

Key finding: The rule-based baseline strongly over-predicts *neutral*, missing 90.5% of negative and 76.8% of positive sentences. Contextual models (e.g. FinBERT) would perform substantially better.

Three documented financial events analysed: daily adjusted close prices retrieved via **yfinance** and cached locally. Window: last trading close before event (T–1) to close five trading sessions after (T+5).

Return formula: `(Price_end – Price_start) / Price_start`. Benchmark: SPY (SPDR S&P 500 ETF).

Event	Date	Ticker	Window Start→End	Ticker Return	SPY Return	Excess Return
SVB Collapse — Bank Run Starts SPDR Regional Banking ETF	2023-03-09	**KRE**	2023-03-08 → 2023-03-16	**-19.91%**	**-0.70%**	**-19.21%**
Credit Suisse Liquidity Crisis Financial Select Sector SPDR	2023-03-15	**XLF**	2023-03-14 → 2023-03-22	**-2.77%**	**+0.48%**	**-3.25%**
UK Mini-Budget Market Crash iShares MSCI United Kingdom ETF	2022-09-23	**EWU**	2022-09-22 → 2022-09-30	**-5.52%**	**-4.55%**	**-0.96%**

Sources: Wikipedia articles for each event (URLs recorded in `results/market_data/event_returns.csv`). Prices from Yahoo Finance via yfinance. Cached in `results/market_data/price_cache.csv`.

■ ***These are historical observations, not causal proof.*** *Multiple confounding factors affect prices over any 5-day window. These returns do not validate the illustrative portfolio shock percentages used in the stress-test prototype.*

Impact, Limitations & Next Steps

Honest assessment and forward path

What This Prototype Demonstrates

- End-to-end pipeline from unstructured text to portfolio risk output
- Rule-based NLP evaluated against 3,453 real labelled sentences
- Observed market returns for 3 documented financial events
- 36 automated unit tests with GitHub Actions CI
- Clean separation of synthetic shocks from real-data evidence

Honest Limitations

- Sentiment recall: only 9.52% for negative class (rule-based baseline)
- Portfolio shocks are illustrative assumptions — not calibrated to data
- Event dates are pre-selected; no live news ingestion
- Historical returns show correlation only — not causal validation
- Synthetic portfolio; no real positions or confidential data used

Realistic Next Steps

NLP Upgrade	Replace keyword rules with FinBERT or a fine-tuned transformer for domain-accurate sentiment
Shock Calibration	Estimate shock distributions from historical event studies rather than fixed assumptions
Live Data Feed	Connect to real-time news APIs (e.g. NewsAPI, Bloomberg) with deduplication
Broader Events	Expand event taxonomy with more granular geopolitical and sector-specific triggers
Portfolio Scale	Generalise to arbitrary portfolios with full position data ingestion

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